Multi-Backbone Ensemble Learning for Robust Breast Cancer Classification on the Mini-DDSM Mammography Dataset

Sheikh Noor Abdullah[1], Nafis Shahriar[1], Mehejabin Tabassum Lamia[1]
Mysha Ibnat Mourin[1], Farah Ulfat Urmi[1], A.B.M. Kabir Hossain[1]
[1]*Department of Information and Communication Engineering*
[1]*Bangladesh Army University of Engineering & Technology*
[1]Natore, Bangladesh
Email: sheikhnoorabdullah02@gmail.com, nafisshahriar60@gmail.com,
mehejabintabassumlamia@gmail.com, myshaibnat15@gmail.com,
farahulfaturmi@gmail.com, kabir111192@gmail.com

***Abstract*—Reported performance on mammography can look stronger than it really is when related images appear in both the training and evaluation partitions. To reduce that risk, we re-evaluate Mini-DDSM using explicit patient grouping, an identifier audit, and repeated grouped cross-validation. The resulting protocol contains 2,417 conservative patient groups from 9,684 images: 682 Normal, 836 Benign, and 899 Cancer groups. Three raw identifiers (A_1500, A_1511, and A_1632) occur in more than one class folder, and four within-class identifiers contain two metadata-distinct four-image blocks. When the released filenames do not provide a stronger validated subject key, those ambiguous records are kept together rather than split across folds. EfficientNet-B4 and ConvNeXtV2-Tiny are trained independently, evaluated with four-view test-time augmentation (TTA), and combined using a fixed 50/50 probability average. The three-fold StratifiedGroupKFold experiment is repeated with seeds 42, 123, and 2026. Across the three repeats, the ensemble obtains a mean patient-group ROC-AUC of 0.8912 ± 0.0076, PR-AUC of 0.8304 ± 0.0120, and default-threshold accuracy of 0.8120 ± 0.0010. At threshold 0.5, sensitivity is 0.6188 and specificity is 0.9264; train-side Youden thresholds raise sensitivity to 0.7875 while specificity becomes 0.8388. We also examine calibration, decision-curve behavior, false-negative cases, and matched-TTA baselines. The emphasis is on a reproducible evaluation protocol rather than on ranking results obtained under incompatible split strategies.**

***Index Terms*—Breast cancer classification, mammography, deep learning, ensemble learning, EfficientNet, ConvNeXt, patient-grouped cross-validation, Mini-DDSM, test-time augmentation, calibration.**

I. Introduction

Breast cancer remains the most commonly diagnosed cancer among women worldwide. GLOBOCAN reported roughly 2.30 million new cases in 2020 [1], 2.30 million in 2022 [2], and 2.43 million in 2024 [3]; Fig. 1 places those published estimates side by side. Mammography is central to breast-cancer screening, but benign and malignant findings can share visual characteristics, and interpretation is influenced by reader variability and workload. Deep-learning methods can support image assessment [4], yet their reported accuracy depends heavily on how the data are divided. If different images from the same patient appear in both training and test sets, the test set is no longer fully independent. For Mini-DDSM, patient grouping is therefore a core part of the evaluation design rather than a minor preprocessing detail.

The revision concentrates on the parts of the pipeline that matter most for reproducibility. We audit the released identifiers before building any fold, verify that patient-group keys never cross the training/test boundary, and repeat the full three-fold experiment with three random seeds. EfficientNet-B4 and ConvNeXtV2-Tiny are assessed separately, as a fold-selected single-best baseline, and as a fixed equal-average ensemble under the same TTA procedure. We also report operating-point metrics, calibration, decision curves, false-negative patterns, and qualitative Grad-CAM examples so that the strengths and limitations of the system are visible beyond a single headline score.

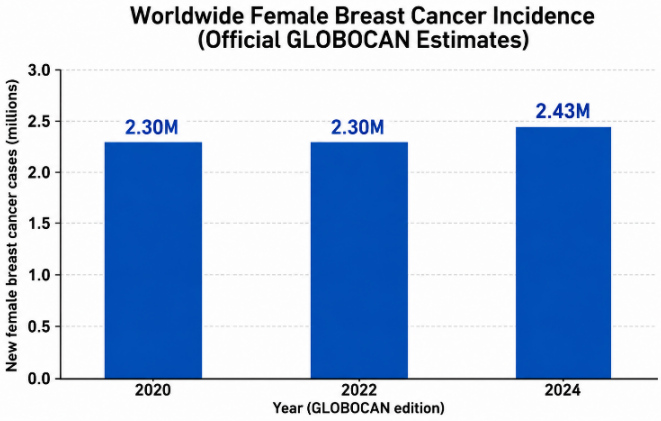


Fig. 1. Worldwide female breast-cancer incidence reported in the 2020, 2022, and 2024 GLOBOCAN editions [1]–[3]. Only the values published in those editions are shown; intermediate years are not interpolated.

The rest of the paper is organized as follows. Section II reviews related mammography work and explains why published image-level accuracies are not directly comparable. Section III

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describes the data audit, grouped validation, preprocessing, models, and evaluation protocol. Section IV presents the quantitative results and qualitative explainability analysis. Section V discusses the main trade-offs and error patterns, and Section VI summarizes the conclusions and limitations.

II. Related Work

Earlier computer-aided mammography systems typically combined hand-crafted texture or shape descriptors with conventional classifiers. The widespread use of ImageNet-pretrained convolutional networks later made transfer learning a practical alternative for medical-image analysis [13]. Table I summarizes representative results reported for Mini-DDSM and closely related DDSM-derived datasets.

Those numbers require some care when interpreted. Several studies in Table I do not state clearly whether every image from a patient was kept in one partition. Their reported accuracies are therefore useful as background, but they are not direct equivalents of the patient-grouped evaluation used here. The review by Fajrin and Min [9] also highlights substantial variation in mammography validation practice. For that reason, Table I provides context rather than a performance ranking.

The two backbones were selected because they make different architectural choices. EfficientNet-B4 couples compound scaling with efficient convolutional blocks [14], whereas ConvNeXtV2-Tiny uses larger kernels, layer normalization, and Global Response Normalization [15]. AlexNet [10], VGG [11], ResNet [12], and ImageNet pretraining [13] provide the broader transfer-learning background. Rather than assuming that model diversity will help, we test that question directly by comparing the fixed equal average against each backbone and against a fold-selected single-best baseline under identical TTA.

TABLE I
Prior Studies on Mini-DDSM / DDSM-Derived Data

Study	Year	Method	Metric
Heenaye-Mamode Khan et al. [5]	2021	ResNet-50, fine-tuned	Acc. 0.88
Sahu et al. [6]	2023	Hybrid CNN (AlexNet+ResNet+MobileNetV2)	Acc. 0.9917
Ansar et al. [7]	2020	MobileNet transfer learning	Acc. 0.87
Khourdifi et al. [8]	2024	VGG16+DenseNet121+InceptionV3 ensemble	Acc. 0.895

Note: Patient-level grouping is not reported consistently in these studies, so the listed accuracies are included for context rather than as direct comparisons with the present protocol.

III. Methodology

A. Dataset Description

Mini-DDSM [16] contains 9,684 grayscale mammograms: 2,728 Normal, 3,360 Benign, and 3,596 Cancer images. For the binary task, Normal and Benign are combined as Not-Cancer. Applying the conservative grouping rule gives 2,417 patient groups: 682 Normal, 836 Benign, and 899 Cancer groups. Figure 2 shows the image-level class counts. Most groups contain four image records, although the released view labels do not always form a complete left/right CC/MLO set. Four within-class identifiers contain eight images and are handled separately in Section III-B. The filename examination token is 1 for every image, so it cannot be used to distinguish longitudinal examinations. A SHA-256 audit identified 14 exact-duplicate pairs; every pair stays within one patient group, and none crosses a group boundary.

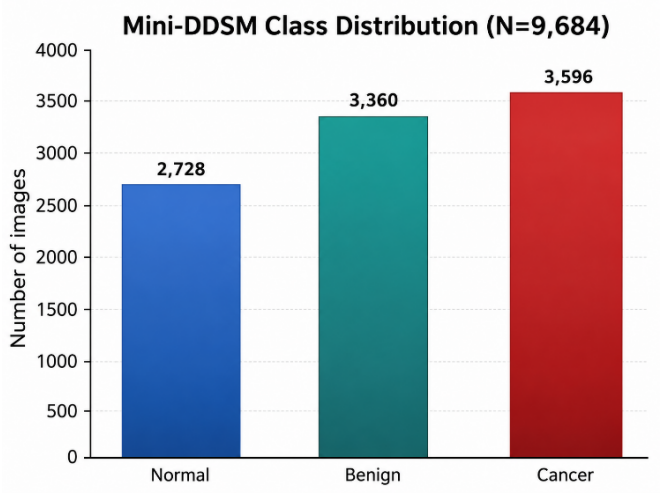


Fig. 2. Image counts in the Normal, Benign, and Cancer folders of Mini-DDSM ($N = 9{,}684$).

B. Patient Identifier Correction

The identifier audit revealed two kinds of ambiguity. First, A_1500, A_1511, and A_1632 occur in both the Benign and Cancer folders, covering 24 image records. To prevent unrelated records from being merged, the grouping key is created as `class_folder::raw_patient_id` before any split is generated; for example, `Benign::A_1500` and `Cancer::A_1500` remain separate. Second, A_1512, B_3470, C_0412, and C_0478 each contain eight images that the metadata separate into two four-image blocks. The released filenames do not provide a stronger validated subject key for safely dividing those blocks, so each raw identifier is kept as one conservative group. This choice may merge distinct subjects, but it avoids placing unresolved records from the same identity in different folds. The source package includes the corrected manifest, ambiguity audits, exact seed-wise splits, leakage checks, and compact pseudocode for the grouping procedure.

C. Patient-Grouped Cross-Validation

We use three-fold StratifiedGroupKFold and repeat the complete experiment with seeds 42, 123, and 2026. Cancer versus Not-Cancer defines the stratification target, while the corrected patient-group key keeps each group intact. Every outer fold contains 805–806 groups, including 299–300 Cancer and 505–507 Not-Cancer groups, so the class proportion changes very little from one fold to another. Three folds keep each held-out set reasonably large while making repeated nested training feasible for two 512×512 backbones. Five- or ten-fold repetition would provide a denser estimate of fold-to-fold

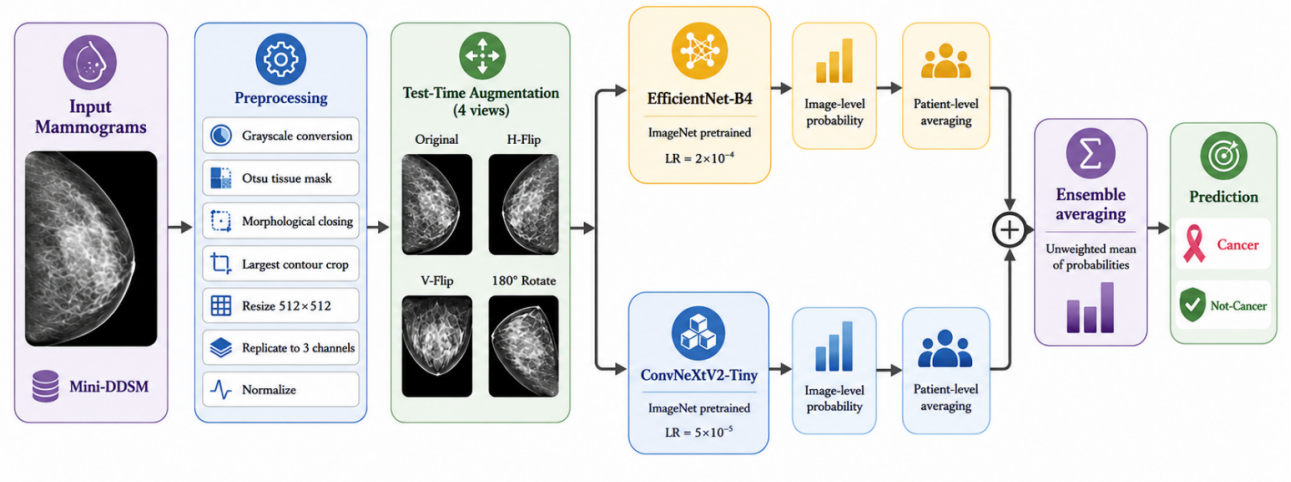


Fig. 3. Inference workflow used for the outer-test comparisons. Each mammogram is evaluated with four TTA views in both backbones. TTA probabilities are first averaged within each backbone, image-level probabilities are then aggregated within the patient group, and the two patient-group scores are combined using a fixed 50/50 average.

variation, but at the cost of smaller test folds and many more training runs. Every split is checked programmatically, and no patient-group key appears in both the outer-training and outer-test partitions.

D. Image Preprocessing

The same preprocessing is applied throughout the study. Otsu thresholding [18] is used to separate breast tissue from the background, the largest tissue contour is cropped, and the image is resized to 512×512 pixels. The single grayscale channel is then replicated three times to match the input format expected by ImageNet-pretrained backbones. Random augmentation is restricted to the inner-training subset; inner-validation and outer-test images are not stochastically modified.

E. Backbone Architectures

Both networks start from ImageNet-pretrained weights and receive the same preprocessed inputs. EfficientNet-B4 [14] is fine-tuned with a learning rate of 2×10^{-4}, while ConvNeXtV2-Tiny [15] uses 5×10^{-5}. AdamW [17] is used for optimization in both cases. EfficientNet-B4 is trained for at most 25 epochs with early-stopping patience 6; ConvNeXtV2-Tiny is capped at 17 epochs with patience 4.

F. Test-Time Augmentation and Ensemble Strategy

At inference, every mammogram is evaluated in four forms: the original image, a horizontal flip, a vertical flip, and a 180° rotation. The four probabilities are averaged within each backbone. Image-level probabilities are then averaged over all images assigned to the same patient group, yielding one group-level score per backbone. The primary ensemble is the arithmetic mean of those two scores, with fixed 0.5/0.5 weights and no tuning on outer-test data. The complete sequence is shown in Fig. 3.

G. Evaluation Protocol

The outer-test fold is kept untouched during model selection. Within each outer-training partition, patient groups are divided into inner training and validation subsets. Checkpoints, fold-specific Youden thresholds, and Platt scaling are selected using only inner-validation predictions and are frozen before the outer test is evaluated. Random augmentation is limited to inner training, and every backbone and ensemble comparison uses the same TTA procedure. Metrics are calculated only after image predictions have been aggregated to the patient-group level. We report seed-level ROC-AUC and PR-AUC, threshold-specific metrics, within-seed patient-group bootstrap intervals, and mean $\pm$ SD across seeds. Where a 95% t-interval is reported across the three seed means, it is treated as a descriptive measure of repeat-split variation rather than as independent-cohort uncertainty.

IV. Results and Analysis

A. Per-Fold Training Results

Table II summarizes the inner-validation AUC values used only for checkpoint selection. Across the nine seed-fold runs, EfficientNet-B4 reaches a mean best AUC of 0.8970 and ConvNeXtV2-Tiny reaches 0.8864. Each seed column averages its three folds, while the final column summarizes all nine runs. ConvNeXtV2-Tiny varies more across the repeated splits, which is one reason a single seed would give an incomplete picture. Figure 4 shows the individual runs.

TABLE II
Seed-Wise Inner-Validation AUC (Each Seed = 3-Fold Mean; No TTA)

Backbone	Seed 42	Seed 123	Seed 2026	9-run mean $\pm$ SD
EfficientNet-B4	0.9081	0.9056	0.8773	0.8970 ± 0.0195
ConvNeXtV2-Tiny	0.9057	0.8848	0.8687	0.8864 ± 0.0310

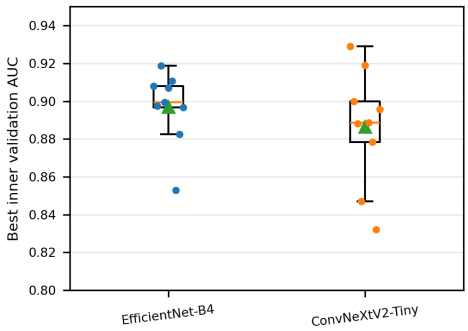


Fig. 4. Best inner-validation AUC across the nine seed-fold runs; values are used only for checkpoint selection.

B. Ensemble Performance

Across the three seeds, the fixed equal-average ensemble reaches a mean patient-group ROC-AUC of 0.8912 ± 0.0076 and PR-AUC of 0.8304 ± 0.0120. The descriptive 95% t-interval for ROC-AUC is 0.8724–0.9101. Mean ROC-AUC is 0.8820 for EfficientNet-B4, 0.8572 for ConvNeXtV2-Tiny, and 0.8775 for the fold-selected single-best baseline. Under the same grouped protocol and TTA, the fixed equal average therefore provides the strongest mean discrimination among the tested fixed strategies without learning ensemble weights. Figure 5 shows the corresponding ROC curves, while Table IV gives the matched operating-point comparison.

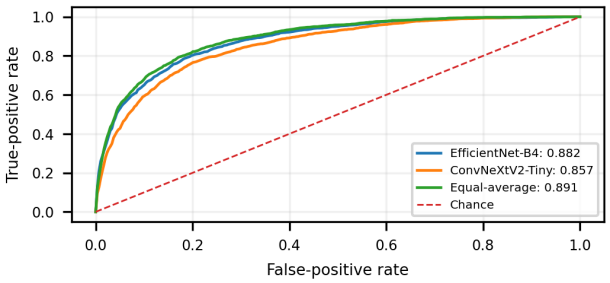


Fig. 5. Mean patient-group ROC curves over the three repeated seeds. Legend values are the corresponding mean seed-level AUCs.

C. Threshold-Based Evaluation

The ranking metrics do not determine the operating point, so Table III reports the threshold-specific results. Point estimates are seed means, and the values in parentheses span the seed-specific 95% patient-group bootstrap intervals. At threshold 0.5, accuracy is 0.8120, sensitivity 0.6188, specificity 0.9264, PPV 0.8335, and NPV 0.8041. Youden thresholds selected from inner-validation predictions increase sensitivity to 0.7875 and NPV to 0.8696, while specificity becomes 0.8388 and PPV 0.7433; accuracy is 0.8197. The Cancer-group false-negative counts fall from 327, 343, and 358 at threshold 0.5 to 181, 187, and 205 at the corresponding train-side Youden thresholds. Figure 6 summarizes the class-normalized confusion rates, and Fig. 7 later shows the overall operating-point shift.

TABLE III
PRIMARY ENSEMBLE METRICS (MEAN; SEED-SPECIFIC 95% BOOTSTRAP CI ENVELOPE)

Metric	@ 0.5	Train-side Youden
Accuracy	0.812 (.796–.829)	0.820 (.801–.839)
Sensitivity	0.619 (.568–.667)	0.788 (.745–.824)
Specificity	0.926 (.900–.950)	0.839 (.807–.864)
PPV	0.833 (.786–.879)	0.743 (.705–.777)
NPV	0.804 (.780–.828)	0.870 (.845–.890)

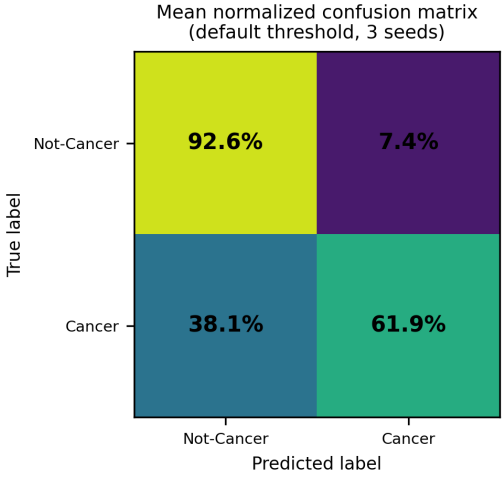


Fig. 6. Class-normalized patient-group confusion matrix averaged across the three seeds at threshold 0.5. Entries are rates rather than pooled sample counts.

D. Calibration and Decision Analysis

Calibration adds information that ROC-AUC alone cannot provide. Platt scaling, fitted only to inner-validation predictions, reduces the ensemble's mean Brier score from 0.1357 to 0.1237 and ECE from 0.0745 to 0.0230. The calibrated values are also lower than the corresponding EfficientNet-B4 values of 0.1312 and 0.0300. In decision-curve analysis, the ensemble shows positive net benefit relative to treat-all and treat-none over thresholds 0.05–0.80 in all three seeds. These results are still internal to Mini-DDSM and should not be interpreted as evidence of clinical effectiveness. Figure 8 presents the calibration curves.

These measures answer questions that ROC-AUC does not. ROC-AUC describes how well cases are ranked across thresholds, while the Brier score and ECE indicate how closely the predicted probabilities align with observed outcomes. Decision-curve analysis adds an operating perspective by asking whether acting on the model could provide more net benefit than treat-all or treat-none across a range of threshold assumptions. Platt scaling is learned only from inner-validation predictions, so outer-test labels remain reserved for evaluation. Even with that separation, the calibration and decision-curve findings should be viewed as internal evidence until they are reproduced on an independent cohort.

E. Ablation Study

Table IV compares EfficientNet-B4, ConvNeXtV2-Tiny, the fold-selected single-best baseline, and the fixed 50/50 ensemble under identical outer-test TTA. Every value is averaged over the same three repeated seeds.

TABLE IV
MULTI-SEED BACKBONE, SINGLE-BEST, AND EQUAL-AVERAGE COMPARISON

Metric	EffNet-B4	ConvNeXt	Single-best	Equal avg.
ROC-AUC	0.8820	0.8572	0.8775	**0.8912**
PR-AUC	0.8198	0.7821	0.8078	**0.8304**
Accuracy	0.8071	0.7777	0.7931	**0.8120**
Sensitivity	**0.6515**	0.5517	0.5832	0.6188
Specificity	0.8992	0.9115	0.9174	**0.9264**
PPV	0.7938	0.7900	0.8071	**0.8335**

The equal-average ensemble has the highest mean ROC-AUC, PR-AUC, accuracy, specificity, and PPV in Table IV, while EfficientNet-B4 remains more sensitive at threshold 0.5. In other words, the ensemble improves overall discrimination but does not dominate every operating metric. Its mean ROC-AUC also exceeds the fold-selected single-best baseline (0.8912 versus 0.8775) even though the ensemble weights remain fixed at 0.5/0.5.

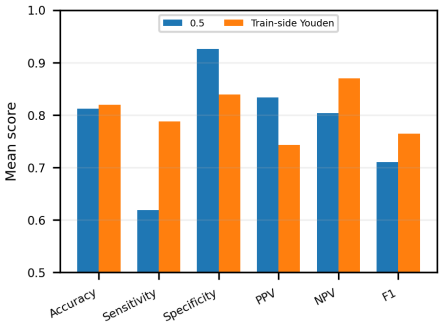


Fig. 7. Mean patient-group operating metrics at the fixed 0.5 threshold and at the fold-specific Youden thresholds selected from training-side validation predictions.

F. Model Explainability Analysis

Grad-CAM [19] is used only as a qualitative inspection tool. The examples in Fig. 9 come from the original model analysis and are retained to show where the network placed attention in one true positive, true negative, false positive, and false negative case. They are not part of the revised multi-seed quantitative evidence and should not be interpreted as validated lesion localization.

The heatmaps are localized rather than uniformly diffuse, but that observation is descriptive. No overlap metric was calculated against radiologist annotations, and Grad-CAM may respond to texture, borders, or preprocessing artifacts as well as pathology. We therefore use it as a transparency aid rather than as quantitative evidence of clinically meaningful localization.

V. DISCUSSION

Repeating the evaluation across three seeds gives a more dependable picture than a single favorable split. The equal-average ensemble improves mean ROC-AUC and PR-AUC over either backbone, but the gain is modest and the operating threshold changes the practical behavior. At 0.5, the model favors specificity and PPV; the train-side Youden threshold recovers more Cancer groups while accepting lower specificity. The published accuracies in Table I remain useful context, but they are not direct benchmarks because patient grouping and split construction are not reported consistently across studies.

False negatives are particularly important in a screening-oriented setting. At threshold 0.5, 327–358 of the 899 Cancer groups are missed across seeds, whereas the train-side Youden thresholds reduce that range to 181–205. Among the 340 Cancer groups missed in all three repeats at 0.5, 137 contain all four distinct left/right CC/MLO combinations and 203 do not. Non-standard view composition is common among persistent misses, but it does not explain why an individual case is misclassified. Without radiologist review or lesion-level annotations, a morphological explanation would be speculative. The system should therefore be treated as a candidate decision-support or second-reader model that still requires external and clinical validation.

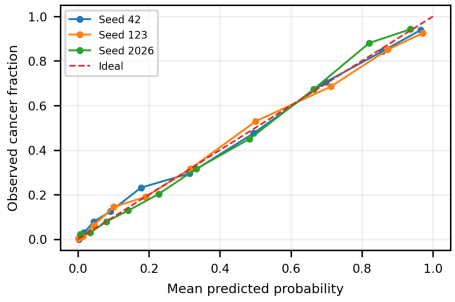


Fig. 8. Patient-group calibration curves after Platt scaling fitted only on inner-validation predictions; one curve is shown for each repeated seed.

VI. CONCLUSION

This study evaluates a fixed EfficientNet-B4/ConvNeXtV2-Tiny ensemble on Mini-DDSM using collision-aware patient grouping and repeated random seeds. Checkpoint selection, threshold choice, and calibration are based only on training-side validation data, leaving each outer-test fold for final evaluation. Across the three seeds, the equal-average ensemble reaches a mean ROC-AUC of 0.8912, PR-AUC of 0.8304, and default-threshold accuracy of 0.8120. Its main value is not dominance at every operating point, but a simple untuned improvement in mean discrimination under one consistent protocol.

The study remains limited to Mini-DDSM. Three folds keep the test partitions large and balanced, but more folds would describe split variability more finely. Four within-class eight-image identifiers are kept intact because the released filenames do not provide a stronger validated subject key, and the examination token cannot separate longitudinal studies. Grad-CAM also remains qualitative. Future work should apply the same evaluation rules to an independent cohort such as CBIS-DDSM or INbreast, use richer metadata where available

to resolve ambiguous identities, and include radiologist review before clinical interpretation.

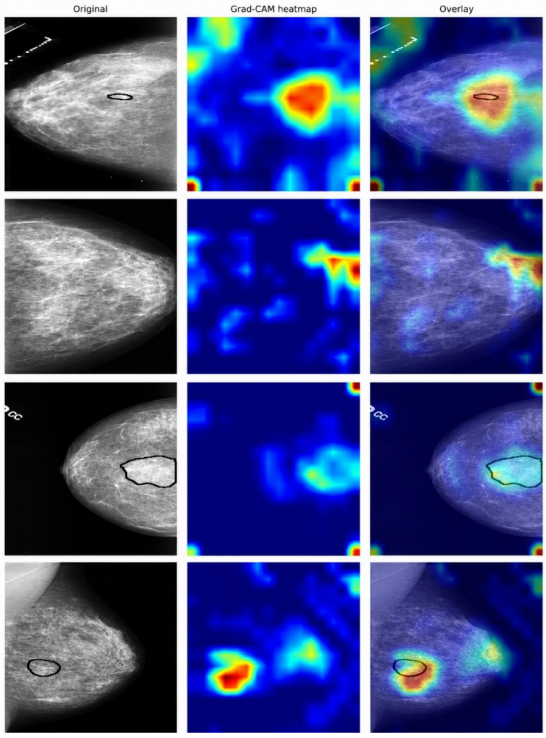


Fig. 9. Qualitative Grad-CAM examples retained from the original model analysis. Columns show (a) the mammogram, (b) the Grad-CAM heatmap, and (c) the overlay; rows show a true positive, true negative, false positive, and false negative. These examples are illustrative and remain separate from the revised multi-seed quantitative analysis.

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